

# Dongfang Zhao, Ph.D.

Website: [faculty.washington.edu/dzhao](http://faculty.washington.edu/dzhao)

## Professional Preparation

|                                           |                                  |      |
|-------------------------------------------|----------------------------------|------|
| University of Washington, Seattle         | Computer Science, PostDoc        | 2017 |
| Illinois Institute of Technology, Chicago | Computer Science, PhD            | 2015 |
| Emory University, Atlanta, GA             | Computer Science, MS             | 2008 |
| Katholieke Universiteit Leuven, Belgium   | Mathematical Statistics, Diploma | 2005 |

## Employment

|              |                                                                           |
|--------------|---------------------------------------------------------------------------|
| 2023–present | Assistant Professor, University of Washington                             |
| 2020–2023    | Lead Developer, Moka Blox LLC, Davis, CA                                  |
| 2018–2020    | Visiting Professor, University of California, Davis (UC Davis)            |
| 2018–2023    | Assistant Professor, University of Nevada, Reno (UNR)                     |
| 2015         | Research Intern, IBM Almaden Research Center, San Jose, CA                |
| 2014         | Research Intern, Argonne National Laboratory, Chicago, IL                 |
| 2013         | Research Intern, Pacific Northwest National Laboratory, Seattle, WA       |
| 2009–2011    | Software Developer, Epic Systems Corporation, Madison, WI                 |
| 2003–2004    | Research Intern, Interuniversity Microelectronics Centre, Leuven, Belgium |

## Selected Awards

|      |                                                                                                                                                                          |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2022 | Federal Research Grant, UNR PI, NSF 2112345, \$255,916<br>“A DLT Machine Learning Platform for Blockchain Warehousing”                                                   |
| 2020 | Federal Research Grant, UNR PI, DOE SC0020455, \$200,000<br>“HPChain: Energy-Efficient Scalable Blockchains for HPC Systems”                                             |
| 2019 | Best Paper Award, International Conference on Cloud Computing<br>“BlockLite: Toward Accurate and Efficient Emulation of Public Blockchains in the Cloud”                 |
| 2018 | Best Student Paper Award, IEEE International Conference on Cloud Computing<br>“Toward Cost-Effective Memory Scaling in Clouds: Symbiosis of Virtual and Physical Memory” |
| 2016 | Postdoctoral Fellowship, Sloan Foundation, \$155,000                                                                                                                     |

## Selected Services

|              |                      |                                                         |
|--------------|----------------------|---------------------------------------------------------|
| 2023–present | Associate Editor     | Journal of Big Data (Springer Nature)                   |
| 2021–2023    | Editorial Board      | IEEE Transactions on Distributed & Parallel Systems     |
| 2018–2022    | Grant Review Panels  | NSF CISE IIS, NSF CISE OAC, DOE ASCR                    |
| 2022         | Search Committee     | One Associate Professor of Databases, UNR               |
| 2022         | Search Committee     | One Assistant Professor of Artificial Intelligence, UNR |
| 2021         | Search Committee     | Two Assistant Professors of Cybersecurity, UNR          |
| 2019         | Organizing Committee | ACM International Middleware Conference, UC Davis       |

## Selected Publications

(**Dongfang Zhao** is the student's <sup>\*</sup>advisor, <sup>†</sup>thesis committee member, or <sup>‡</sup>postdoc co-author)

- [1] <sup>\*</sup>Olamide Timothy Tawose, Jun Dai, Lei Yang, **Dongfang Zhao**. Toward Efficient Homomorphic Encryption for Outsourced Databases through Parallel Caching. Proceedings of the ACM on Management of Data, SIGMOD 2023.
- [2] <sup>\*</sup>Olamide Timothy Tawose, <sup>\*</sup>Bin Li, Lei Yang, Feng Yan, **Dongfang Zhao**. Topological Modeling and Parallelization of Multidimensional Data on Microelectrode Arrays. The 36th IEEE International Symposium on Parallel and Distributed Processing (IPDPS), 2022. Direct acceptance
- [3] <sup>\*</sup>Abdullah Al-Mamun, Feng Yan, **Dongfang Zhao**. SciChain: Blockchain-enabled Lightweight and Efficient Data Provenance for Reproducible Scientific Computing. 37th IEEE International Conference on Data Engineering (ICDE), 2021.
- [4] <sup>\*</sup>Abdullah Al-Mamun, Feng Yan, **Dongfang Zhao**. BAASH: lightweight, efficient, and reliable blockchain-as-a-service for HPC systems. The International Conference for High Performance Computing, Networking, Storage and Analysis (SC), 2021. Direct acceptance without revision
- [5] <sup>\*</sup>Xinying Wang, <sup>\*</sup>Olamide Timothy Tawose, Feng Yan, **Dongfang Zhao**. HDK: Toward High-Performance Deep-Learning-Based Kirchhoff Analysis. The Thirty-Fourth AAAI Conference on Artificial Intelligence (AAAI), 2020. Oral presentation
- [6] <sup>†</sup>Heyang Qin, <sup>†</sup>Syed Zawad, Yanqi Zhou, Lei Yang, **Dongfang Zhao**, Feng Yan. Swift machine learning model serving scheduling: a region based reinforcement learning approach. The International Conf. for High Performance Computing, Networking, Storage and Analysis (SC), 2019.
- [7] <sup>‡</sup>Parmita Mehta, <sup>‡</sup>Sven Dorkenwald, **Dongfang Zhao**, Tomer Kaftan, Alvin Cheung, Magdalena Balazinska, Ariel Rokem, Andrew J. Connolly, Jacob VanderPlas, Yusra AlSayyad. Comparative Evaluation of Big-Data Systems on Scientific Image Analytics Workloads. Proceedings of the VLDB Endowment, VLDB 2017.
- [8] **Dongfang Zhao**, Kan Qiao, Jian Yin, Ioan Raicu. Dynamic Virtual Chunks: On Supporting Efficient Accesses to Compressed Scientific Data. IEEE Transactions on Services Computing (TSC). 9(1): 96-109 (2016).
- [9] **Dongfang Zhao**, Ning Liu, Dries Kimpe, Robert B. Ross, Xian-He Sun, Ioan Raicu. Towards Exploring Data-Intensive Scientific Applications at Extreme Scales through Systems and Simulations. IEEE Transactions on Parallel and Distributed Systems (TPDS). 27(6): 1824-1837 (2016).
- [10] **Dongfang Zhao**, Li Yang. Incremental Isometric Embedding of High-Dimensional Data Using Connected Neighborhood Graphs. IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI). 31(1): 86-98 (2009).

## Doctoral Students (name, graduation year, first employment)

|                        |      |                                                       |
|------------------------|------|-------------------------------------------------------|
| Olamide Timothy Tawose | 2023 | Assistant Professor, Lincoln University, PA           |
| Abdullah Al-Mamun      | 2022 | Assistant Professor, Augusta University, GA           |
| Xinying Wang           | 2022 | Research Associate, Oak Ridge National Laboratory, TN |